

## Introduction

Hidden Markov Models (HMMs) are widely used to characterize dynamic functional connectivity (dFC) in resting-state and task fMRI [1]. Due to the high dimensionality of fMRI data, regularization methods are commonly applied to avoid overfitting. Principal Component Analysis (PCA) reduces dimensionality by projecting data into a lower-dimensional space [2], while Graphical Lasso (GL) imposes sparsity on the inverse covariance matrix [1]. However, PCA makes covariance recovery in the original space difficult, and GL can introduce state imbalance, where infrequent states yield unstable predictions. In addition, GL regularization is naturally formulated within Expectation–Maximization (EM) frameworks, where the M-step remains convex with exact solutions, whereas incorporating the same constraints into gradient-based discriminative training complicates optimization and breaks the original GL formulation. In this abstract, we present an alternative way to regularize fMRI data using a mixture of factor analyzers (MFA) [3].

## Dataset

### ➤ 7T Pattern separation [4]

- 46 Cognitively Normal (CN) and 41 Mild Cognitive Impairment (MCI).
- 137 CN scans and 119 MCI scans.
- TR=1.53, time points = 510.

### ➤ Preprocessing

- (1) Slice-timing correction and rigid-body realignment.
- (2) Co-registration.
- (3) Spatial normalization to MNI152.
- (4) Detrending.
- (5) Summation within the CONN 32 mask.
- (6) Temporal normalization to zero mean and unit variance.

### ➤ Variables

fMRI data:  $X \in \mathbb{R}^{N \times T \times p}$       Number of Hidden states:  $K$   
Number of scans:  $N = 256$       Number of features:  $p = 32$   
Number of time points:  $T = 510$

## Regularization

### ➤ Gaussian

$$\mu_k \in \mathbb{R}^{p \times 1} \quad \sigma_k \in \mathbb{R}^{p \times p}$$

Number of independent parameter:  $Kp + Kp(p+1)/2$

### ➤ Gaussian + GL

$$\mu_k \in \mathbb{R}^{p \times 1} \quad \Theta_k \in \mathbb{R}^{p \times p}$$

$$\Theta_k = \arg \min_{\Theta_k} [-\log \det(\Theta_k) + \text{tr}(\sigma_k \Theta_k) + \lambda \|\Theta_k\|_1]$$

Number of independent parameter:  $Kp + \text{Non-zero elements in } \Theta$

### ➤ Gaussian + PCA

$$\mu_k \in \mathbb{R}^{q \times 1} \quad \sigma_k \in \mathbb{R}^{q \times q} \quad U \in \mathbb{R}^{p \times q}$$

$$X_{st} W \sim \mathcal{N}(\mu_k, \sigma_k)$$

Number of independent parameter:  $pq + Kq + Kq(q+1)/2$

### ➤ Gaussian + MFA

$$\mu_k \in \mathbb{R}^{q \times 1} \quad W_k \in \mathbb{R}^{p \times q} \quad \Psi_k \in \mathbb{R}^{p \times 1}$$

$$X_{st} \sim \mathcal{N}(\mu_k, W_k W_k^T + \text{diag}(\Psi_k))$$

Number of independent parameter:  $Kp + Kpq + Kp$

## Acknowledge

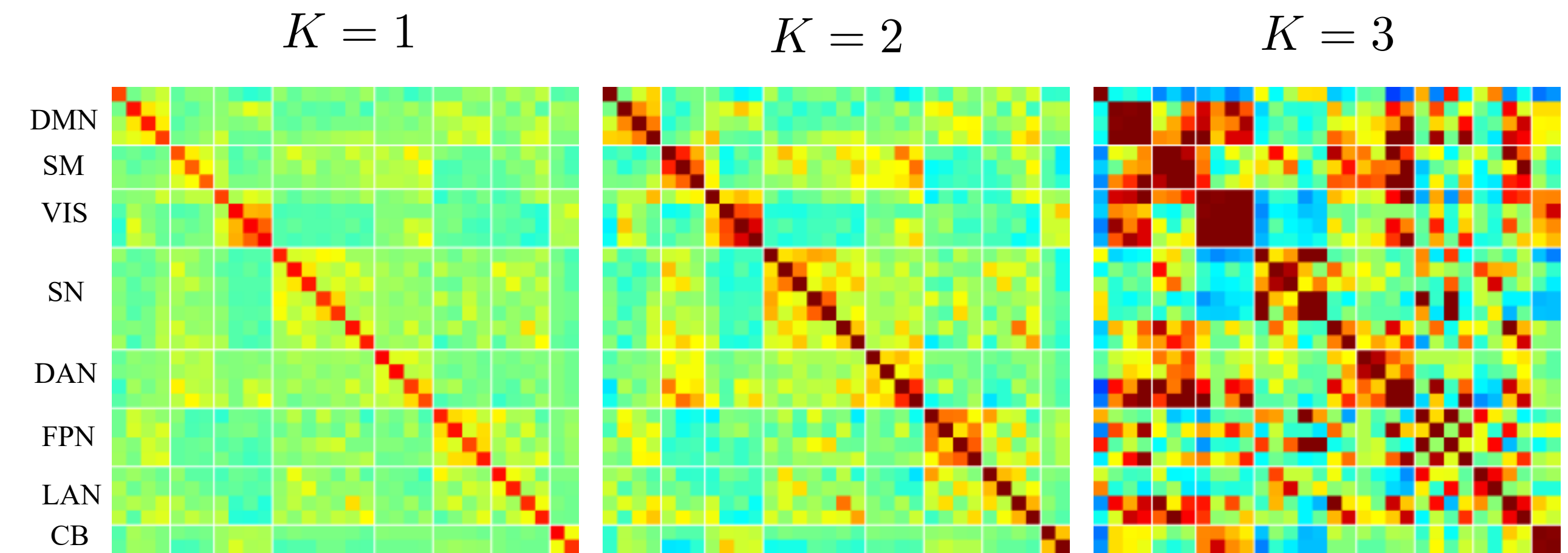
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## Reference

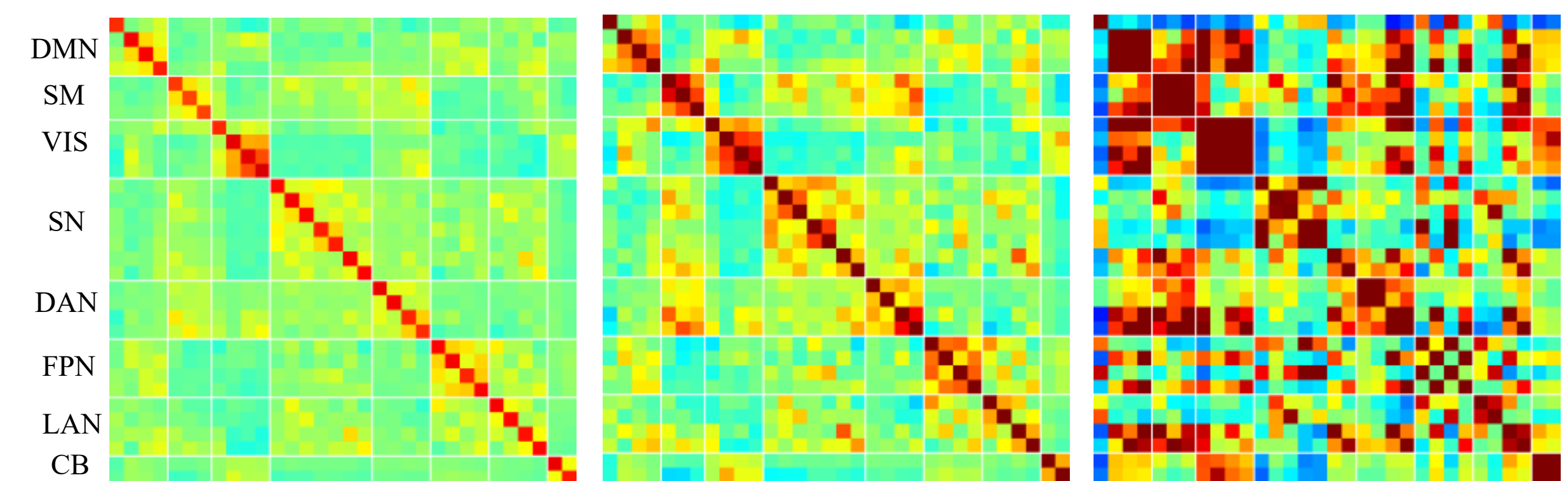
- [1]. Zhang G, Cai B, Zhang A, Stephen JM, Wilson TW, Calhoun VD, Wang YP. Estimating dynamic functional brain connectivity with a sparse hidden Markov model. IEEE Trans Med Imaging. 2019;39(2):488–498.
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- [5]. Shappell H, CaUo BS, Pekar JJ, Lindquist MA. Improved state change estimation in dynamic functional connectivity using hidden semi-Markov models. Neuroimage. 2019;191:243–257.

## Results

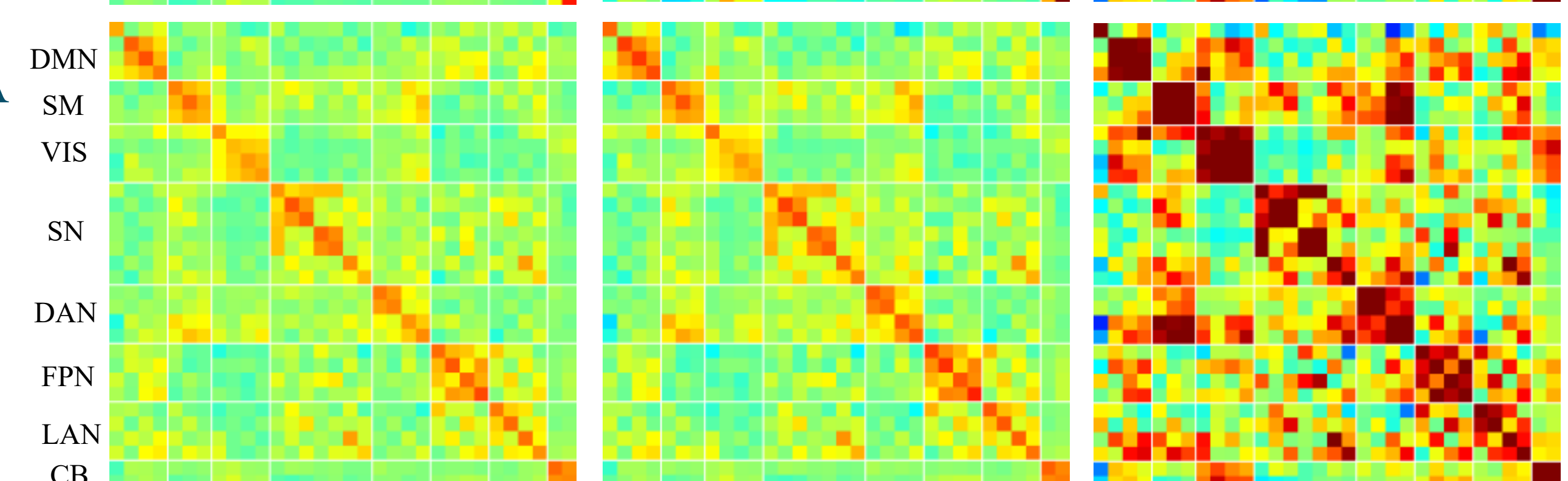
### ➤ Gaussian



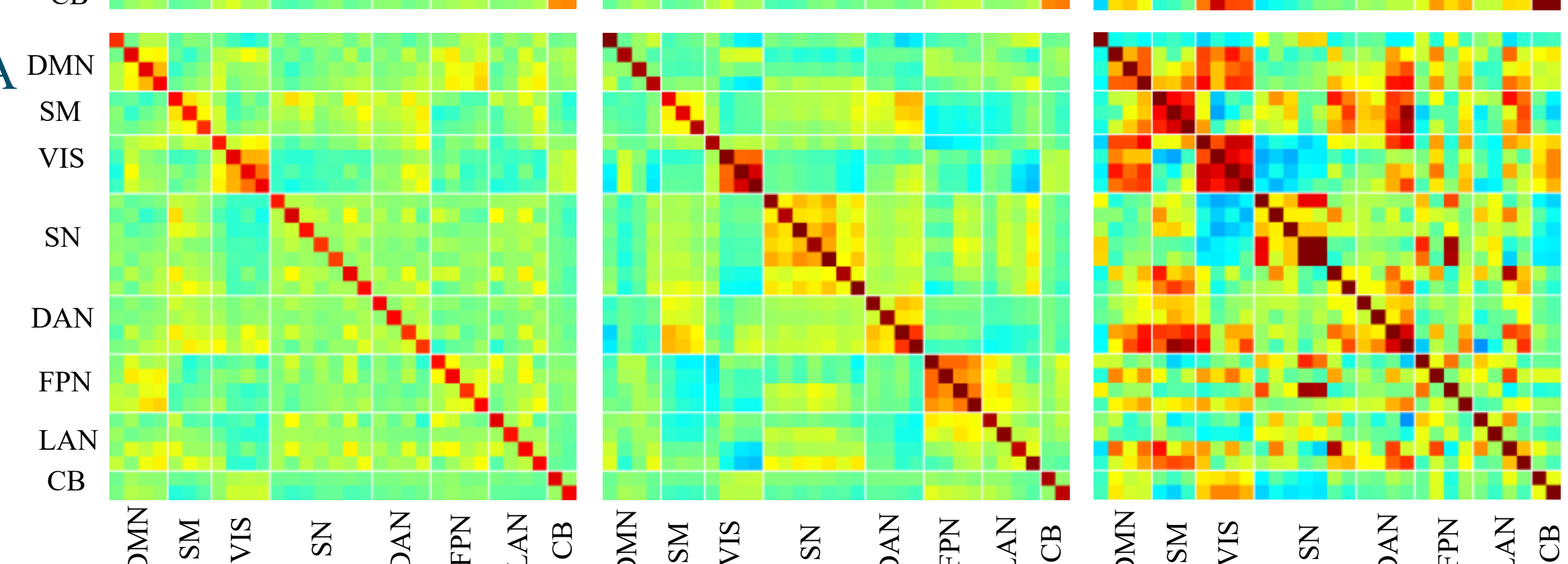
### ➤ Gaussian + GL



### ➤ Gaussian + PCA



### ➤ Gaussian + MFA



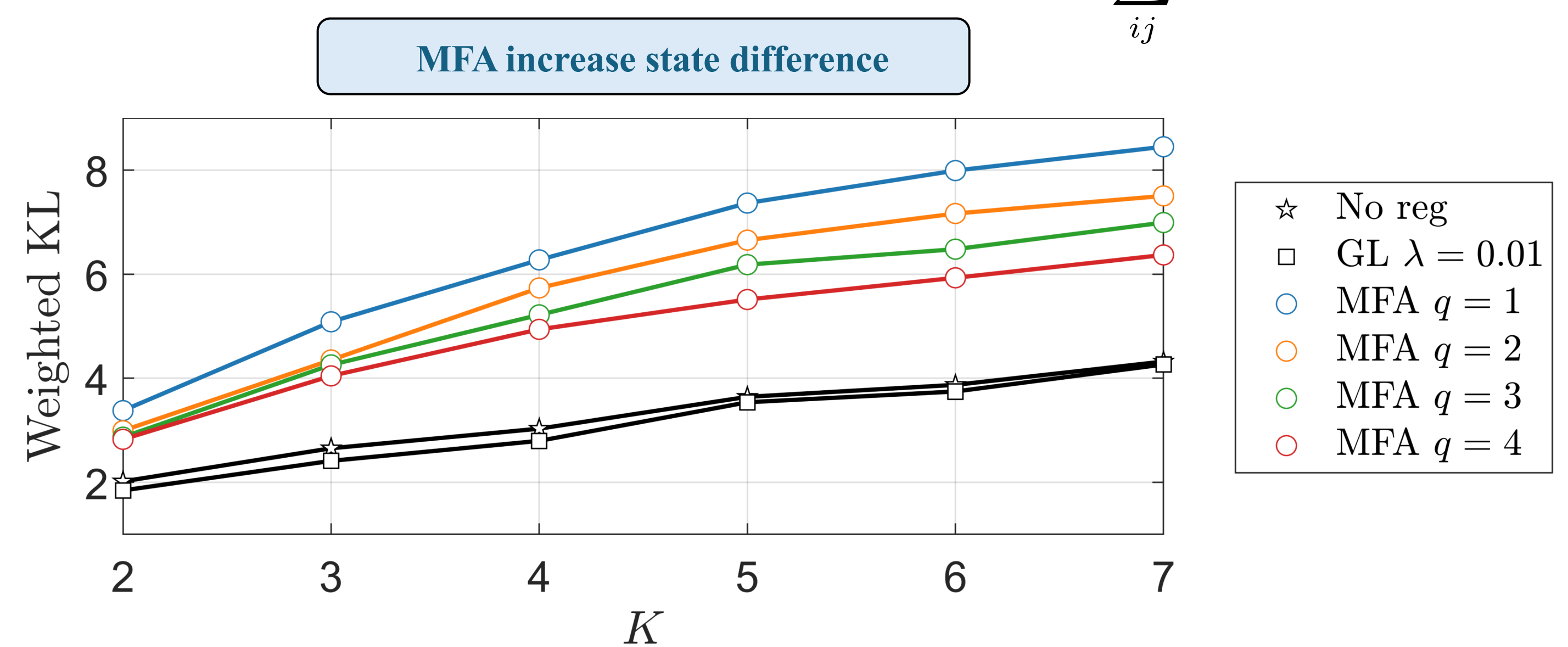
### ➤ Properties

Unbalanced state observed for GL

| Method         | Number of independent parameter | Occupational probability |       |       |
|----------------|---------------------------------|--------------------------|-------|-------|
|                |                                 | K=1                      | K=2   | K=3   |
| Gaussian       | 1680                            | 0.513                    | 0.423 | 0.065 |
| Gaussian + GL  | 1394                            | 0.644                    | 0.313 | 0.043 |
| Gaussian + PCA | 654                             | 0.444                    | 0.384 | 0.172 |
| Gaussian + MFA | 576                             | 0.493                    | 0.349 | 0.158 |

### ➤ Weighted Kullback-Leibler (KL) divergence:

$$\max \sum_{ij} p_i p_j \text{KL}_{ij}.$$



### ➤ BIC [5]:

MFA reduces BIC

$$\min -2 \log(p) + \text{Num par} \times \log(NT).$$

